

Seven gene sets. One auditable curation.

A concise record of the publication rules, pathway selection, identity checks, and overlap that produced the stocker-ready inputs.

7

stocker CSVs

2

analysis families

4

primary sets

3

GO term sets

Two curation families—analyzed separately

Primary evidence sets and CSW-derived GO term-grouped sets use separate overlap analyses.

FAMILY 1 • PRIMARY CURATION SETS

01

6

genes

Mechanistic

Publication core + screen
ranks 1–6

02

20

genes

Homeostatic H×R

15 History+/Rebound– + 5
opposite

03

97

genes

CSW 4+ (FC0.5)

FC0.5 wake frequency ≥ 4

04

4

genes

HLH regulators

Four bHLH factors named in
paper

FAMILY 2 • CSW-DERIVED GO TERM-GROUPED SETS (FC0.5 / FC1)

05

99

genes

Ribosomal

DAVID term hits; approved
keywords

06

167

genes

Mitochondrial

DAVID mito/metabolism
term hits

07

5

genes

Immune

DAVID immune / Toll / Imd
term hits

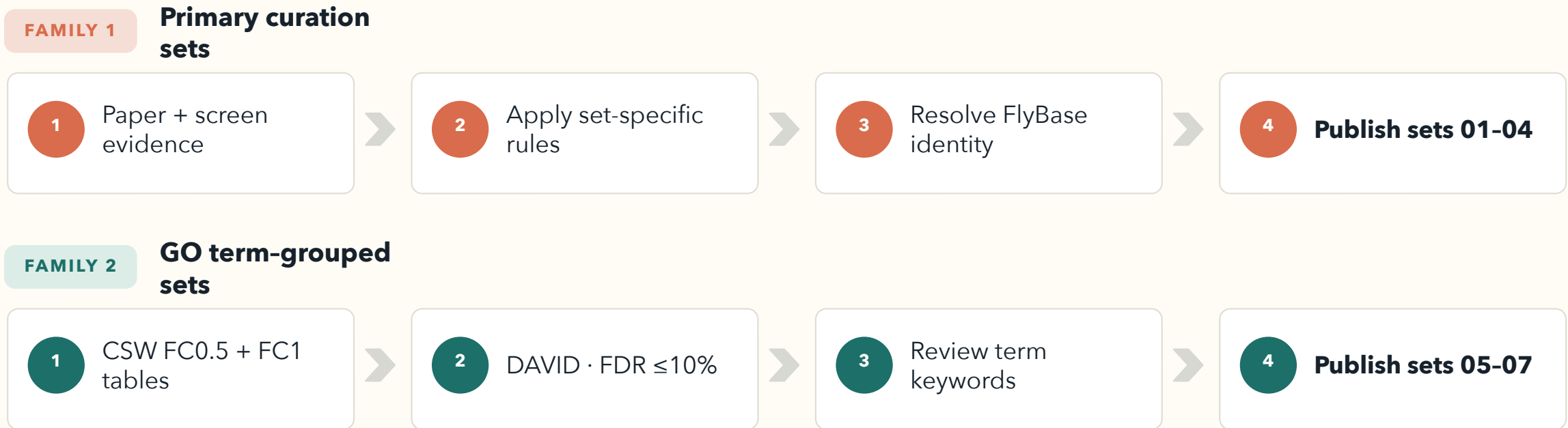
RULE

Keep them separate

Report overlap within each
family—never as one seven-set
comparison.

Two curation tracks converge on one controlled folder

They share identity gates and file standards—but their overlap analyses remain separate.

**902**

source observations

791

unique source FBgns

 ≥ 2

minimum pathway frequency

7

experiments

Thresholds remain separate: FC0.5 and FC1 are never collapsed into a max-frequency field.

Publication sets use explicit, reproducible rules

Sets 01-02 preserve publication evidence and resolve one biologically important identity conflict.

SET 01 · 6 GENES

Mechanistic

Publication core joined by FBgn to mechanistic screen ranks 1-6.

- Carry forward mechanism, literature status, Tx-Omics evidence, and fl.ai confidence.
- Evidence labels: consistent, correlated, or curated.
- No rank or annotation is inferred outside the source tables.

unc79 · SIFa · rumpel · AstA-R2 · Trhn · RpL23

SET 02 · 20 GENES

Homeostatic history × rebound

15

History+ / Rebound-

5

History- / Rebound+

Identity correction

Source row trh / FBgn0262139 is resolved from publication context to Trhn / FBgn0035187 (tryptophan hydroxylase). The wrong FBgn is never retained.

Screen frequency and direct-regulator evidence stay separate

Set 03 is threshold-scoped screen evidence; Set 04 comes directly from the paper.

SET 03 · 97 GENES

CSW 4+ wake-frequency union

97

FC0.5 wake frequency 4-6

7

also qualify at FC1

0

sleep genes at frequency 4

Union rule

- Take FC0.5 wake genes at frequency 4, 5, or 6.
- Add FC1 wake frequency-4 genes; all seven are already in the FC0.5 set.
- Keep FC0.5 and FC1 metadata in separate columns.

SET 04 · 4 GENES

HLH regulators

Named in Results / Table S1 after HOMER identified a CAGCTG E-box upstream of wake-induced genes.

bigmax FBgn0039509

HLH3B FBgn0011276

E(spl)m3-HLH FBgn0002609

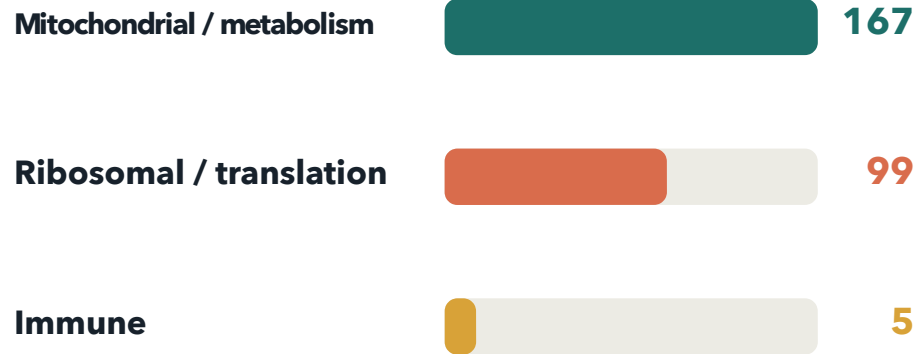
E(spl)mbeta-HLH FBgn0002733

CSW-derived GO term-grouped genes retain threshold provenance

Each gene records its source as FC0.5, FC1, or both; only hits from approved FDR-passing terms enter sets 05-07.

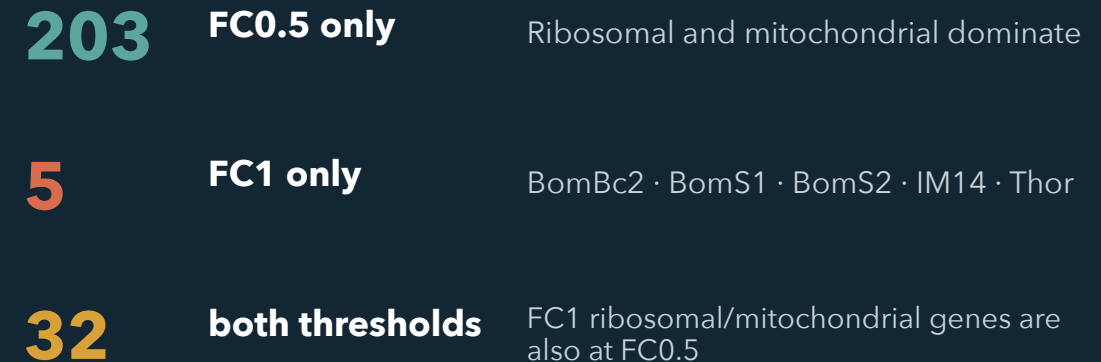


GO term-grouped outputs



All three outputs are wake-direction genes.

Threshold provenance · 240 unique FBgns



Key takeaway Only the five immune genes are FC1-exclusive.

Identity resolution is conservative by design

Current symbols can pass directly; ambiguous synonyms require biological context or a case-supported source match.

PASS**Trhn****FBgn0035187**

Current symbol · tryptophan hydroxylase neuronal · publication-resolved in Homeostatic.

REJECT**trh****FBgn0262139**

Current symbol · trachealess · must not enter any generated identity field for this publication row.

REVIEW**Trh****ambiguous**

Synonym of Hn, Trhn, and trh · never auto-approved without biological context.

Approval gates

- Exact current symbol
- Publication-supported correction
- Unique case-insensitive current-symbol match
- Otherwise: stop for review

Membership key

flybase_gene_id

Symbols are display labels. All overlap, deduplication, and set membership use FBgn only.

Primary-set overlap is limited to three genes

Mechanistic, Homeostatic, CSW 4+ (FC0.5), and HLH are compared only with one another.

6

Mechanistic

20

Homeostatic

97

CSW 4+ · FC0.5

4

HLH

Observed shared genes

Mechanistic × Homeostatic**2**

Trhn · AstA-R2

Mechanistic × CSW 4+ (FC0.5)**1**

RpL23

All other pairings**0**

No shared FBgns

Within Family 1

124

unique FBgns

121

in exactly one set

3

in exactly two sets

0

in three or four sets

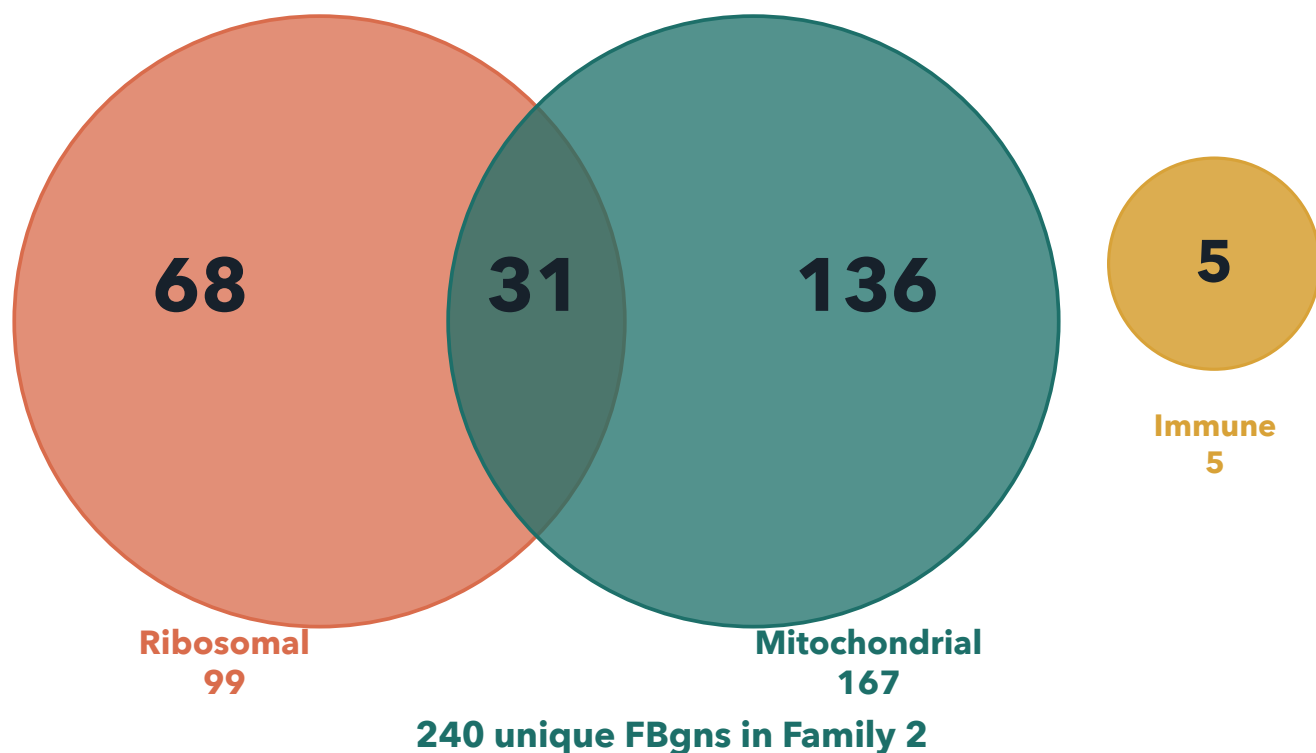
HLH is fully disjoint from the other primary sets.

GO term-grouped overlap is confined to ribosomal × mitochondrial

The three pathway buckets are compared only within Family 2; threshold provenance remains a separate attribute.

GO term-group membership

Schematic · counts shown



What the overlap means

31 genes occur in both ribosomal and mitochondrial buckets

5 immune genes are fully disjoint

0 immune overlaps with either other GO bucket

The 31-gene overlap is intentional: mitochondrial translation terms match both keyword groups.

The contract is simple—and strict.

7

Exactly seven CSVs
in the discovery
folder

2

Required keys:
ext_gene and
flybase_gene_id

0

No overlap, GO, or
audit CSVs beside
the inputs

RUN

```
python -m fl_ai_reagent_stocker run ./data/gene_sets/Tx-Omics_Revision/ \
--config ./data/config/stock_split_config_priority_example.json
```

Seven inputs

- 01 Mechanistic
- 02 Homeostatic HxR
- 03 CSW 4+
- 04 HLH regulators
- 05 Ribosomal
- 06 Mitochondrial
- 07 Immune

Stability rule

Discovery is recursive: every extra *.csv changes the run. Keep analysis outputs elsewhere.